Siddhesh Amrale

Senior Software Engineer | Data & AI | Full-Stack Cloud
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SUMMARY

Azure-certified engineer (AI-102, AZ-204, AZ-104) with 5+ years building data & AI platforms, full-stack applications, and reliable cloud services processing 500M+ events/day. Delivered $400K+/yr in cost savings, architected multi-source ingestion frameworks, shipped production React applications, built Azure OpenAI RAG agents, and drove reliability through SLI/SLO practices across distributed cloud services.

TECHNICAL SKILLS

Data Platforms: Databricks (Spark/PySpark), Azure Data Explorer (KQL), Delta Lake, Apache Iceberg, Apache Airflow, dbt, Snowflake, Apache Kafka
Data Engineering: ETL/ELT Pipeline Design, Dimensional Data Modeling (Star/Snowflake Schema, SCD Type 2), Streaming & Batch Processing, Event-Driven Architecture, Incremental Loads (Watermark/CDC), Data Governance, Data Quality
AI & ML: Azure OpenAI, Azure AI Search, Azure Machine Learning, AWS Bedrock, RAG (Retrieval-Augmented Generation), Vector Search, Embeddings, Semantic Kernel, Prompt Engineering, MLflow, AI Agents
Frontend: React, Angular, TypeScript, HTML/CSS, Redux, WebSockets, Jest, Cypress
Cloud & Infrastructure: Azure (Functions, Event Grid, IoT Hub, Event Hub, Blob Storage, Key Vault, API Management, DevOps), AWS (EC2, S3, Lambda, EKS), Docker, Kubernetes (AKS, EKS), Terraform, GitHub Actions, ArgoCD, CI/CD, Prometheus, Grafana, OpenTelemetry, SLIs/SLOs/SLAs
Languages: SQL/KQL, Python, C#, Java, JavaScript/TypeScript, Node.js, C++, .NET, FastAPI, Express.js, Spring Boot, GraphQL, REST
Databases: PostgreSQL, MySQL, Redis, MongoDB, Azure Cosmos DB, Azure SQL, Elasticsearch

PROFESSIONAL EXPERIENCE

Senior Software Engineer Jan 2024 - Present
Allegheny County Airport Authority, Pittsburgh, PA

- **Multi-Source Ingestion Framework:** Architected a reusable ingestion platform on Azure Functions supporting REST APIs, IoT Event Hub streams, Snowflake, and batch file feeds (JSON, CSV, ZIP) with 9 paging strategies, Managed Identity auth, and watermark-based CDC loads; reduced new-domain onboarding from weeks to 2 days and identified $400K+/yr in cost savings by migrating pipelines from ADF to serverless.
- **Medallion Data Pipelines & BI Products:** Built Databricks medallion pipelines (Bronze to Silver to Gold) with MLflow experiment tracking across parking, flights, baggage, and predictive maintenance; applied AI entity-matching with confidence scoring to map source values to gold-standard dimensions, delivering Power BI data products across 5 business units with sub-second analytics on 10+ dashboards.
- **Full-Stack Operational Monitoring Apps:** Designed and delivered multiple full-stack web applications (React, C#, Python) with Cypress E2E test coverage for real-time monitoring of airport concessions, sensors, equipment, and flight operations, backed by live IoT Hub to Event Hub to Azure Data Explorer streaming pipelines; reduced incident response time by 40%.
- **AI RAG Agents & Workflow Automation:** Built RAG agents using Azure OpenAI and Semantic Kernel, orchestrated via Airflow, enabling five business teams to self-serve operational queries in natural language; reduced ad-hoc reporting requests by approximately 60% and cut average report turnaround from days to real-time by surfacing governed data through conversational AI interfaces.

Software Engineer Sep 2022 - Nov 2023
Freddie Mac, New York City, NY

- **Full-Stack Data API Platform:** Designed and built a unified GraphQL + REST data gateway (Node.js, Java, React) with Azure API Management, Entra ID auth, distributed caching, and rate limiting; sustained 1K+ RPS at under 10ms P95 latency serving real-time mortgage decisioning data to internal applications and downstream consumers.
- **Site Reliability Engineering:** Implemented circuit breaker, bulkhead isolation, idempotency, and exponential backoff patterns across distributed data services; defined SLIs/SLOs and managed error budgets, achieving and sustaining 99.95% uptime across production services.
- **Cloud-Native Modernization:** Containerized 12+ microservices on Kubernetes (AKS, EKS) across Azure and AWS infrastructure with canary and blue-green deployments via ArgoCD; established CI/CD pipelines through GitHub Actions cutting release cycles by 50% and reducing deployment-related incidents by 30%.

Software Engineer, Full-Stack Jan 2019 - Jul 2021
Top Hat Software Technologies, Pune, India

- **Full-Stack Application Development:** Built and maintained full-stack web applications (Angular, Java, C++, MongoDB) supporting media content delivery across Azure Blob Storage and Cloud CDN at under 90ms p99 latency, serving 10M+ daily requests; built batch ETL pipelines to ingest and transform content metadata, cutting data refresh lag from 4 hours to under 20 minutes.
- **Platform Reliability & Observability:** Instrumented end-to-end platform observability using Prometheus and Grafana, building dashboards and alerting for latency, throughput, and error rate SLIs; drove platform from 99.5% to 99.9% availability by identifying and resolving reliability bottlenecks proactively.

CERTIFICATIONS

- Microsoft Certified: Azure AI Engineer Associate (AI-102) | View Credential
- Microsoft Certified: Azure Developer Associate (AZ-204) | View Credential
- Microsoft Certified: Azure Administrator Associate (AZ-104) | View Credential

EDUCATION

M.S. Computer Science | New Jersey Institute of Technology Sep 2021 - May 2023
B.E. Computer Science | Pune University Jul 2015 - May 2019